ELSEVIER

Talanta 50 (1999) 743–749

Talanta

www.elsevier.com/locate/talanta

TOC versus UV spectrophotometry for wastewater quality monitoring

O. Thomas [a,*], H. El Khorassani [a], E. Touraud [a], H. Bitar [b]

[a] *Laboratoire Génie de l'Environnement Industriel, Ecole des Mines, 6 avenue de Clavières 30319, Alès Cedex, France*
[b] *Laboratoire, Faculté des Sciences, avenue Normandie Niemen, 13013, Marseille, France*

Abstract

Total organic carbon (TOC) is one of the most important parameter for the knowledge of water and wastewater quality, because it concerns theoretically all organic compounds. Unfortunately, some restrictions with respect to TOC measurement must be considered, explaining that alternative procedures have been envisaged, among which UV spectrophotometry. Starting from a comparison of results between high temperature digestion and UV photo-oxidation techniques for some specific compounds and real wastewater samples, the work shows the complementary interest of using UV spectrophotometry either directly (with multiwavelength procedures) or after UV photo-oxidation. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: TOC; UV spectrophotometry; Wastewater; Monitoring

1. Introduction

Total organic carbon (TOC) is the most relevant parameter for the global determination of organic pollution of water and wastewater. It has been proposed in the 1970s namely for automatic survey because of problems related to the field use of BOD and COD. Actually, the organic carbon is used for that purpose or for the completion of the knowledge of organic pollution, as old parameters quantify its main effect, the oxygen consumption. It is the reason why TOC is often considered as the 'true' parameter. Unfortunately, some problems exist with the use of TOC measurement for wastewater due to technical or metrological consideration. TOC measurement seems to give a specific information (the organic carbon content of the sample) but only leads to a global aggregate response.

The aim of this work is to test the TOC availability from several experiments on wastewater and to propose a complementary approach, based on the use of UV spectrophotometry, in order to complete the knowledge of wastewater quality and to bring some explanation to the composition of organic matrix.

2. TOC availability

TOC measurement can be performed by using two main techniques for the conversion of organic

* Corresponding author.

PII: S0039-9140(99)00202-7

Table 1
Comparison of TOC values for different analysers

TOC meter type	Number of instruments	Measurement type	Average value of series (ppm)	Mean difference (%)
High temperature	2	Laboratory	37.2	8.5
High temperature	2	Laboratory	354.1	16.1
UV persulfate + high temperature	1	Laboratory	318.7	30.2
UV persulfate	2	On line	222.8	26.2

carbon into carbon dioxide. The first one involves a UV photo-oxidation at low temperature with a persulfate solution, after stripping of the mineral fraction, and the second one uses a catalytic oxidation at high temperature (650–900°C). This last technique is chosen for wastewater as the conversion yield is more efficient for anthropogenic organic matter. Both technologies use a NDIR detector for the final measurement of carbon dioxide. Other techniques are proposed either for the oxidation step improvement or for the final determination of carbon dioxide, but these latter lead to more fragile instruments than the basic techniques [1,2]. Unfortunately, as the cold technique is more easy to run, TOC on line analysers (actually off line) are often built around this procedure.

A comparison between the two techniques has been carried out in a petrochemical site. Four TOC analysers, two on site (off line) and two in laboratory have been tested. The analysers were identical for each purpose (cold technique — Seres — off line, and high temperature — Shimadzu — in laboratory) as well as the technical team for experiments and maintenance. Fresh standard solutions used for calibration were also identical. Several samples of raw and treated wastewater, characterised by a low concentration of suspended solids, were simultaneously analysed with two TOC analysers depending on the experiment. More than 20 samples have been considered for each experiment. Table 1 presents the different results and some observations can be done:

- for a same technique, the mean difference between the results obtained from two high-temperature laboratory instruments is 8.5 and 16.1%, respectively for samples with about 40 and 350 mg l^{-1} of TOC. For two UV-persulfate on line analysers, the mean difference is 26.2% for about 220 mg l^{-1} of TOC.
- For the two techniques, the mean difference between two laboratory instruments is 30.2% for about 220 mg l^{-1} of TOC. Fig. 1 shows the distribution of results.

The observed difference between the set of results can be explained by several reasons among which, a best efficiency of high temperature instruments for industrial wastewater containing refractory anthropogenic organic matrix or the heterogeneity of sampling (for on line instruments).

One important point is the availability of the measure, it means the frequency during which the user gets an acceptable result without any trouble of the instrument. For a year and the four instruments of the petrochemical site, the percentage of the mean availability is about 80% of the time with good efforts for maintenance, representing each year about 50% of the investment cost of the analysers.

In conclusion, even if these observations seem to be penalizing for TOC, this parameter remains the most relevant and its measurement is rather fast.

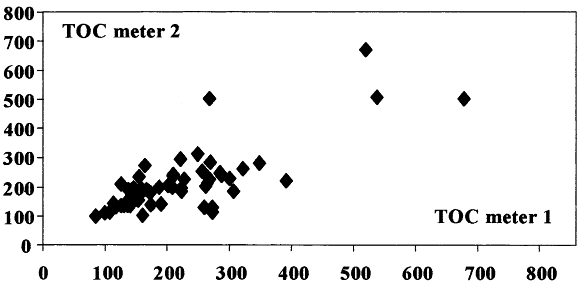

Fig. 1. Dispersion of results, expressed in mg l^{-1} of TOC, for two on line analysers.

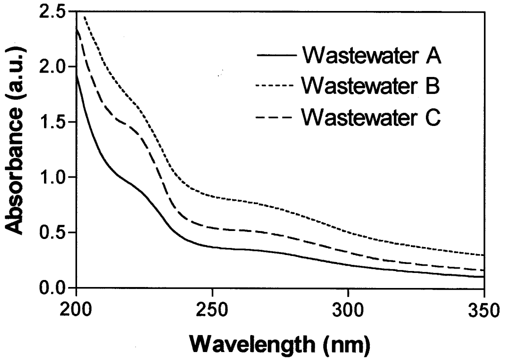

Fig. 2. UV spectra of wastewater samples (TOC = 57 mg l^{-1}).

3. Interest of UV spectrophotometry for TOC explanation

The interest of the explanation of TOC is evident as it is important to know the stability degree of organic matter in water. In raw wastewater, the fresh organic pollution will be degraded, first slowly in sewers and then, more efficiently in a biological treatment plant for example. This evolution is depending on the presence of microorganisms and oxygen. After the treatment plant, the residual TOC is very low and can be accepted by water body.

In some cases, a better knowledge of the organic matrix corresponding to a given value of TOC could be useful. For example, for a given organic pollution load, there is a difference between one wastewater sample from urban origin with a wide variety of organics and another one characterised by the presence of a major pollutant coming from an industrial discharge in sewer. In the last case, the diagnosis of the presence of a potential toxicity is important. On the other hand, two samples with low content of TOC could be either a treated effluent or a diluted raw sample. For the environment, the effects of both water are obviously not similar. Another example given in Fig. 2 shows UV spectra of different wastewater samples with a same value of TOC (57 mg l^{-1}).

The spectra shape can be explained by the difference of composition of the organic matrix in the aqueous, colloidal and solid phases. For this purpose, the COD values would be of poor interest meanwhile the results of BOD could be probably useful. Unfortunately, its experimental conditions are not acceptable for an efficient control. Other parameters as total hydrocarbons, anionic surfactants or phenol index could be used but with a very low probability to give relevant results. Analytical procedures as chromatographic methods could also be considered if samples composition has already be studied. Otherwise, it is not realistic to use separation techniques.

UV spectrophotometry is regularly proposed and used for water and wastewater quality control for 50 years [3], either from urban [4,5] or industrial origin [6,7]. Its main quantitative application concerns the estimation of aggregate parameters (TOC, COD, TSS, anionic surfactants, ...) [8–11] and the determination of specific compounds (nitrate, chromium VI, phenol, ...) [12,13] namely by the robust exploitation of spectra [14,15] using direct multiwavelength procedures [16,17]. Other quantitative applications of UV spectrophotometry concern the estimation of PAH in soils [18] or of biodegradable dissolved organic carbon (BDOC) [19]. The use of UV spectrophotometry is possible because of the basic interaction between UV light and unsaturated ionic or molecular structures (chromophores). Moreover a simple UV photo-oxidation step can be proposed for the estimation of reduced nitrogen forms (TKN and ammonia). In this technique, called UV/UV, a final UV determination of nitrate is included [20].

On the other hand, UV spectrophotometry can give relevant qualitative information as for example for the study of treated wastewater discharges [21], for the detection of incidents in industrial context [7] or for the treatability study of industrial wastewater [22].

Table 2 presents the TOC and UV responses of several organic compounds families.

The study of TOC and UV responses of organic compounds shows obviously that saturated molecules do not absorb even in the UV region. Moreover, small molecules like simple carboxylic acids or aldehydes absorb in the far UV region (< 200 nm), showing only the tail of their spectra. TOC is generally more efficient for organic compounds characterisation but can be limited for the study of some molecules. The detection of these latter can be envisaged either directly (S compounds) or after photo-oxidation (ketones, amines) with UV spectrophotometry.

Before considering some applications for wastewater quality monitoring, an attempt to compare the advantages and drawbacks of the two techniques can be proposed (Table 3). From this comparison, it is obvious that the two techniques are complementary.

Table 2
TOC and UV responses of organic compounds[a]

Compounds	TOC	UV	UV/UV
Saturated compounds	Y	N	–
Aliphatic unsaturated hydrocarbons	Y	P	–
Aromatic compounds	Y	Y	–
Acids	Y	P	–
Aldehydes, ketones	P	P	Y
Alcohols	Y	N	Y
Phenolic compounds	Y	Y	–
Aliphatic amines	P	N	Y
Aromatic amines	Y	Y	–
N unsaturated heterocycles	Y	Y	–
S unsaturated heterocycles	P	Y	–
Humic like substances	P	Y	

[a] Y: 90–100% of conversion or high absorption or absorption after photo–oxidation; P: partially converted or some absorbing compounds; N: non absorbing compounds.

Table 3
Comparison of TOC and UV spectrophotometry: advantages and drawbacks

	TOC	UV
Advantages	Aggregate parameter	Simple, fast, not expensive
	Reference	Qualitative information
	Quick measurement	Other parameters (NO_3^-, CrVI, . . .)
Drawbacks	Some refractory organics	Saturated hydrocarbons
	Suspended solids	Low MW compounds
	Measure availability	Sensitivity, selectivity but dilution often needed
	Chloride	

4. Applications

4.1. Variations of raw wastewater quality

TOC monitoring is often used for the monitoring of industrial wastewater quality. Fig. 3 presents the TOC variation with time (during 3 months) monitored at the inlet of a wastewater treatment plant of a petrochemical site. For each TOC measurement (every 3 h), a UV spectrum has been acquired. Considering that the maximum admitted value for the protection of the biological reactor of the plant is 500 mg l^{-1} of TOC, three peaks of pollution have been detected.

The study of the corresponding UV spectra and a comparison with the one of regular wastewater shows that these three incidents are related to three different accidental discharges in sewer. Moreover, the study of UV spectra shows that incidents 1 and 3 are characterised by the presence of a few concentrated pollutants. Starting from this information, and after a more complete study, a UV monitor has been installed close to the main source of the potential accidental discharge.

4.2. Treatment plant control

TOC and UV measurements can also be used for the control of wastewater treatment plant. If the treatment efficiency is the major goal of an industrial wastewater treatment plant in term of TOC yield for example (Fig. 4), other objectives are more and more considered as the nitrogen removal possibility for urban treatment plant (Fig. 5).

Some complementary observations can be made form the UV spectra shape of Figs. 4 and 5. The closeness of the shape of raw samples spectra is not so evident taking into account the position of the 220–240 nm shoulders and the relative absorbance values around 230 and 270 nm. On the other hand, the spectrum of the outlet 1 (Fig. 5) is characterised by a high residual absorption after 300 nm related to the presence of suspended solids. Moreover, the increase of absorbance values around 210 nm seen on some spectra, is due to the appearance of nitrate in wastewater.

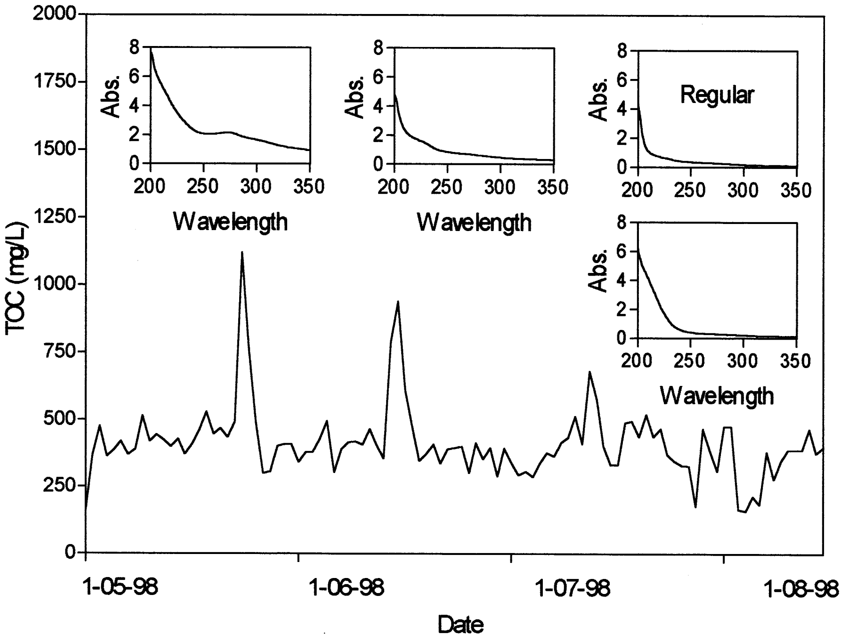

Fig. 3. TOC and UV monitoring of petrochemical wastewater (the absorbance values are for a 1 cm optical pathlength quartz cell).

4.3. Discharge survey

UV spectrophotometry can be used when TOC measurement is very difficult to carry out (high concentration of chloride). Fig. 6 displays the different spectra acquired for a dispersion study of treated wastewater discharge in sea water. The set of spectra shows an isosbestic point due to the mixing of wastewater with sea water, the spectrum of which presenting a sharp absorbance at 205 nm due to chloride (notice that the difference between spectra of nitrate and chloride solutions is chemometrically easy to exploit). After 100 m, the impact of the discharge is difficult to see.

5. Conclusion

The study of characteristics of TOC measurement and UV spectrophotometry shows that both techniques are complementary for the monitoring of wastewater quality. These techniques can be used for regulation applications but also for process control including the detection of incidents (early warning). On line TOC meters and UV monitors (diode array systems) are available for the purpose. Moreover, several developments of UV analysers are interesting to consider in order to complete the on line monitors, such as field portable spectrophotometers, immersed sensors, simple kit for NTK and ammonia estimation.

Thus, the best available solution for wastewater monitoring nowadays is the choice of a TOC analyser equipped with a TSS kit, completed by a

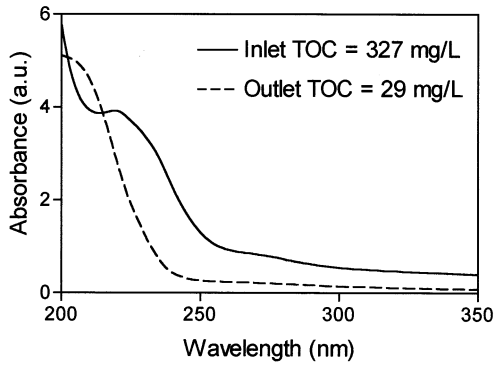

Fig. 4. Treatment plant efficiency for petrochemical wastewater (removal of 91% of TOC — measured — and 85% of TOC — estimated with a UV deconvolution procedure [8]).

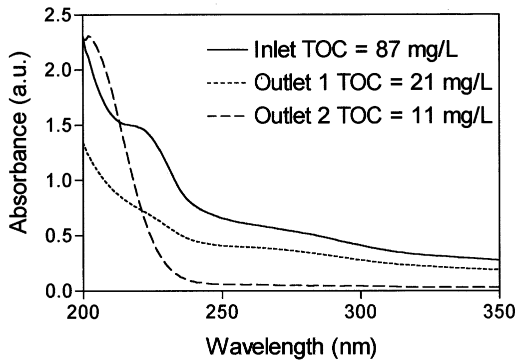

Fig. 5. Efficiency of urban wastewater treatment plant (outlet 1 and 2, respectively without and with nitrification step).

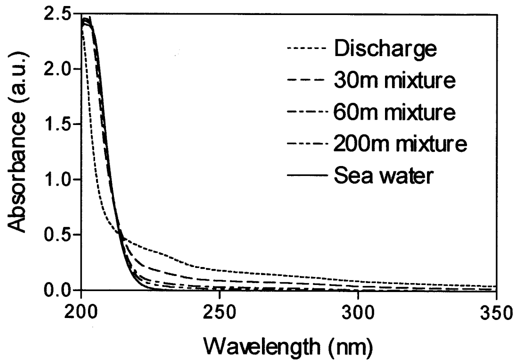

Fig. 6. Discharge of treated industrial wastewater in sea water: set of UV spectra.

UV detector including a NTK kit. In some cases, namely for industrial application, the measurement of conductivity and pH could be useful. If possible an autosampler should be coupled with the chosen on line systems.

Acknowledgements

The authors wish to acknowledge Caroline Lhospitallier and Pierre Trébuchon from Naphtachimie for their contribution.

References

[1] R.W. Matthews, M. Abdullah, G.K.C. Low, Photocatalytic oxidation for total organic carbon analysis, Anal. Chim. Acta 233 (1990) 171–179.

[2] J. Qian, K. Mopper, Automated high-performance, high-temperature combustion total organic carbon analyzer, Anal. Chem. 68 (1996) 3090–3097.

[3] R.C. Hoater, Application of spectrophotometry in the examination of water, Wat. Treat. Exam. 2 (1952) 9–19.

[4] R.A. Dobbs, R.H. Wise, R.B. Dean, The use of ultraviolet absorbance for monitoring the Total Organic Carbon content of water and wastewater, Water Res. 6 (10) (1972) 1171–1180.

[5] M. Mrkva, Automatic U.V.-control system for relative evaluation of organic water pollution, Water Res. 9 (1975) 587–589.

[6] E. Naffrechoux, N. Mazas, O. Thomas, Identification rapide de la composante industrielle d'une eau résiduaire, Envir. Technol. 12 (1991) 325–332.

[7] H. El Khorassani, P. Trébuchon, H. Bitar, O. Thomas, Petrochemical wastewater management assistance, Water Sci. Technol., to be published, 1999.

[8] O. Thomas, F. Théraulaz, C. Agnel, S. Suriany, Advanced UV examination of wastewater, Envir. Technol. 17 (1996) 251–261.

[9] F. Théraulaz, L. Djellal, O. Thomas, Simple LAS determination in sewage using advanced UV spectrophotometry, Tenside Surf. Det. 6 (1996) 447–451.

[10] O. Thomas, F. Théraulaz, V. Cerda, D. Constant, P. Quevauviller, Wastewater quality monitoring, Trends Anal. Chem. 16 (7) (1997) 419–424.

[11] B. MacCraith, K.T.V. Grattan, D. Connolly, R. Briggs, W.J.O. Boyle, M. Avis, Results of a cross-comparison study: optical monitoring of total organic carbon (TOC) of a limited range of samples, Sensors Actuators B 22 (1994) 149–153.

[12] A.R. Hawthorne, S.A. Morris, R.L. Moody, R.B. Gammage, Duvas as a real time, field portable wastewater monitor for phenolics, J. Envir. Sci. Health A19 (1984) 253–266.

[13] H. El Khorassani, G. Besson, O. Thomas, Direct UV visible determination of chromium VI in industrial wastewater, Qumica Anal. 16 (1997) 239–242.

[14] S. Gallot, O. Thomas, State of the art for the examination of UV spectra of water and wastewater, Intern. J. Envir. Anal. Chem. 52 (1993) 149–158.

[15] S. Gallot, O. Thomas, Fast and easy interpretation of a set of absorption spectra: theory and qualitative applications for UV examination of water and wastewater, Fresenius J. Anal. Chem. 346 (1993) 976–983.

[16] O. Thomas, S. Gallot, Ultraviolet multiwavelength absorptiometry, Fresenius J. Anal. Chem. 338 (1990) 234–244.

[17] O. Thomas, F. Théraulaz, M. Domeizel, C. Massiani, UV spectral deconvolution: a valuable tool for wastewater quality determination, Envir. Technol. 14 (1993) 1187–1192.

[18] E. Touraud, M Crône, O. Thomas, Rapid diagnosis of polycyclic aromatic hydrocarbons (PAH) in contaminated soils with the use of ultraviolet detection, Field Anal. Chem. Technol. 2 (1998) 221–229.

[19] O. Thomas, N. Mazas, C. Massiani, Determination of biodegradable dissolved organic carbon in waters with the use of UV absorptiometry, Envir. Technol. 14 (1993) 487–493.
[20] B. Roig, C. Gonzalez, O. Thomas, Measurement of dissolved total nitrogen in wastewater by UV photo-oxidation with peroxodisulfate, Anal. Chim. Acta, accepted for publication.
[21] H. El Khorassani, F. Théraulaz, O. Thomas, Application of UV spectrophotometry to the study of treated wastewater discharges in rivers, Acta Hydrochim. Hydrobiol. 26 (1998) 296–299.
[22] O. Thomas, H. El Khorassani, L. Castillo, UV treatability test for chemical and petrochemical wastewater, Water Sci. Technol., to be published, 1999.